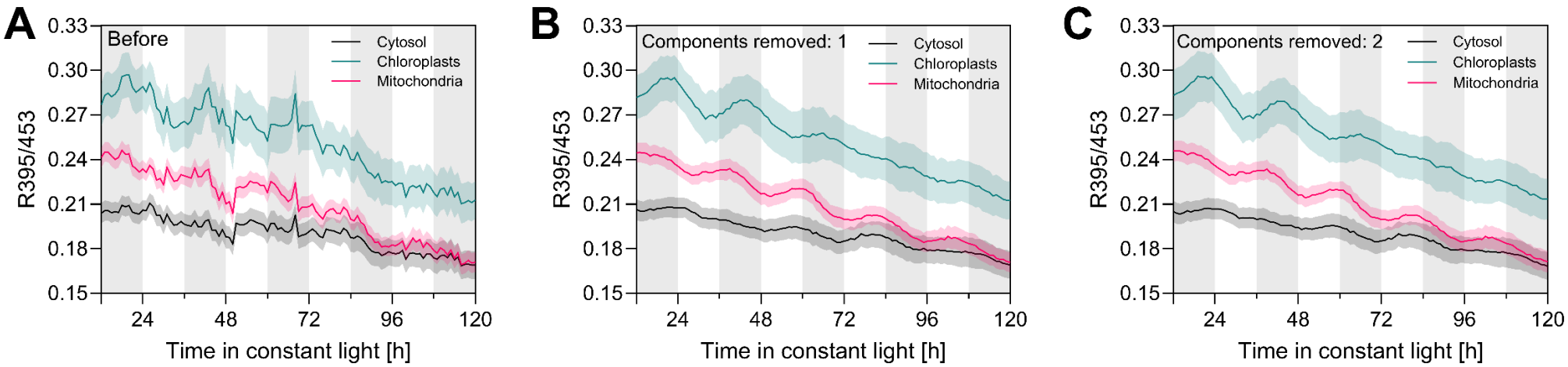


Demonstration of the circadian denoising script on an example dataset. A) Prior to denoising, average signals for the three compartment-localised roGFP2-Orp1 lines (shown as 395/453 nm ratio, cytosol-localised in black, chloroplasts-localised in green, mitochondria-localised in magenta) exhibit a high degree of shared noise due to technical constraints of long-term automated image capture. Based on the homogeneity of the noise observed across the groups, a SVD-based (Singular Value Decomposition) denoising method was employed to identify the shared signal and subtract it from each sample (see the supplemented script). The method was modified to minimize the risk of including any circadian oscillatory pattern as part of the shared signal. The modified version of the script was tested on synthetic datasets aimed to mimic the behaviour of the actual data. The number of shared waveforms (components) to be subtracted from the samples was determined statistically based on similarity in principal component (PC) values between the data subsets. For example, if the groups were found to be statistically different for PCn, then n-1 shared waveforms were to be subtracted. To avoid excessive denoising, the maximum number of subtracted components was kept at 2. The oscillatory pattern for each group remained intact regardless of the number of subtracted components (1 component in (B), 2 components in (C)).